

Math 220B HW 1

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1 D

Problem 1

Prove or disprove that $\mathbb{Z}[x]$ is a Euclidean domain.

Solution: We'll prove that $\mathbb{Z}[x]$ is not a Euclidean domain.

First, recall that any Euclidean domain is a PID, so it suffices to show that $\mathbb{Z}[x]$ is not a PID.

Consider the ideal $(2, x) \subseteq \mathbb{Z}[x]$, we claim that it is not a principal ideal: Suppose the contrary that it is a principal ideal, there exists $f(x) \in \mathbb{Z}[x]$, such that $(f(x)) = (2, x)$. In particular, $f(x) \neq 0$.

This implies that there exists nonzero polynomials $g(x), h(x) \in \mathbb{Z}[x]$, such that $2 = g(x)f(x)$ and $x = h(x)f(x)$.

Now, recall that since $\mathbb{Z}[x]$ is an integral domain, then for any nonzero polynomials $f, g \in \mathbb{Z}[x]$, one has $\deg(f \cdot g) = \deg(f) + \deg(g)$. So, one gets the following:

$$0 = \deg(2) = \deg(g(x)f(x)) = \deg(g(x)) + \deg(f(x)) \quad (1.1)$$

Since degree is always nonnegative for a polynomial, the above implies $\deg(f(x)) = \deg(g(x)) = 0$, hence $f(x) = a$ and $g(x) = b$ for some nonzero $a, b \in \mathbb{Z}$.

Then, apply the same logic to the other equality, one gets the following:

$$1 = \deg(x) = \deg(h(x)f(x)) = \deg(h(x)) + \deg(f(x)) \quad (1.2)$$

Since $\deg(f(x)) = 0$ is proven above, it indicates $\deg(h(x)) = 1$, so $h(x) = cx + d$ for some $c, d \in \mathbb{Z}$ (where $c \neq 0$).

Which, $x = h(x)f(x) = (cx + d)a = (ca)x + (da)$, showing that $ca = 1$ and $da = 0$.

Yet, $ca = 1$ implies that a is a unit, hence $f(x) = a$ is a unit in $\mathbb{Z}[x]$. This concludes that $\mathbb{Z}[x] = (a) = (f(x)) = (2, x)$.

However, notice that $(2, x)$ is a proper ideal: Suppose it is not proper, there exists $k(x), l(x) \in \mathbb{Z}[x]$, such that $2 \cdot k(x) + x \cdot l(x) = 1$. Let's write $k(x) = k_0 + k_1x + \dots + k_nx^n$ and $l(x) = l_0 + l_1x + \dots + l_mx^m$. Then, the equality becomes the following:

$$\begin{aligned} 1 &= 2 \cdot k(x) + x \cdot l(x) = 2(k_0 + k_1x + \dots + k_nx^n) + x(l_0 + l_1x + \dots + l_mx^m) \\ &= 2k_0 + (k_1x + \dots + k_nx^n) + (l_0x + l_1x^2 + \dots + l_mx^{m+1}) \end{aligned} \quad (1.3)$$

Which, the constant term on the left is 1, while the constant term on the right is $2k_0$, showing that $1 = 2k_0$, or 2 divides 1. Yet, this is a contradiction. So, $(2, x)$ must be a proper ideal.

But, if $(2, x)$ is proper, $(2, x) = \mathbb{Z}[x]$ is another contradiction. So, the initial assumption is false, $(2, x)$ can't be a principal ideal.

Since $\mathbb{Z}[x]$ has ideals that're not principal, it's not a PID, hence not a Euclidean domain.

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Problem 2

Give an example of a commutative ring different from $\mathbb{Z}$ and polynomial rings, which is not a PID, but whose every nonzero prime ideal is maximal.

Solution: Consider the ring $\mathbb{Z}_2 \times \mathbb{Z}_2$, which it's commutative, not isomorphic to $\mathbb{Z}$ or any polynomial ring (since both $\mathbb{Z}$ and $R[x]$ over any nonzero ring R are infinite), and it's not a PID (since it's not an Integral domain: one has $(1, 0) \cdot (0, 1) = (0, 0)$ in this ring).

Now, we claim that all of its nonzero prime ideals are maximal. Given any prime ideal $P \subseteq \mathbb{Z}_2 \times \mathbb{Z}_2$, one has $\mathbb{Z}_2 \times \mathbb{Z}_2 / P$ be an integral domain that is finite. However, all finite integral domain must be a field (proof below):

Given R a finite integral domain, then for any nonzero $a \in R$, since $a^n \neq 0$ for all $n \geq 1$ due to the integral domain property, then the collection $\{a, a^2, \dots, a^n, \dots\} \subseteq R$ are consist of nonzero elemens. But, with R being finite, at some point this subset must repeat, so there exists positive integers $i < j$, such that $a^i = a^j$. As a result, $a^i(1 - a^{j-i}) = 0$ (Note: $j - i > 0$); with R being an integral domain and $a^i \neq 0$, this equality enforces $1 - a^{j-i} = 0$, so $a^{j-i} = 1$, showing $a \cdot a^{j-i-1} = 1$, which a is invertible.

Hence, all nonzero prime ideals of $\mathbb{Z}_2 \times \mathbb{Z}_2$ must be maximal, which is an example for this problem.

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Problem 3

Let R be a commutative ring and let $S = R[x]$. Prove or disprove that $U(R) = U(S)$.

Solution: We'll give a counterexample:

Choose $R = \mathbb{Z}_4$, one has $2 \in R$ satisfies $2^2 = 4 \equiv 0 \pmod{4}$, so 2 is a nilpotent element.

Now, consider the element $1 - 2x \in R[x]$, it is a unit, because its square gives the following:

$$(1 - 2x)^2 = (1 - 2x) + (-2x)(1 - 2x) = 1 - 2x - 2x + 2^2x^2 = 1 - 4x + 0x^2 = 1 \quad (3.1)$$

So, $1 - 2x$ is a unit in $R[x]$ (with its inverse being itself), or $1 - 2x \in U(S)$ (with $S = R[x]$).

Yet, $1 - 2x \notin U(R)$ (when viewing R as a subring of $R[x]$) since it's not constant, showing that $U(S) \neq U(R)$ in this case.

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Problem 4

Let R be a commutative ring and $f(x) \in R[x]$ be a zero-divisor. Prove or disprove that there exists $a \in R$ such that $af(x) = 0$ in $R[x]$.

Solution: We'll prove that if $f(x) \in R[x]$ is a zero-divisor, there exists nonzero $a \in R$ such that $af(x) = 0$ in $R[x]$. (Here, one can assume $f(x) \neq 0$, as the case $f(x) = 0$ is trivially true for all nonzero $a \in R$)

Suppose that $f(x) = a_n x^n + \dots + a_0 \in R[x]$ is a zero-divisor, there exists a nonzero $g(x) = b_m x^m + \dots + b_0 \in R[x]$, such that $f(x)g(x) = 0$. In particular, one can choose $g(x)$ to have the minimum (i.e. choose the smallest m , so that there exists polynomial $h(x) \in R[x]$, with $\deg(h) = m$, and $f(x)h(x) = 0$).

First, notice that $f(x)g(x) = 0$ implies its leading term given by $a_n b_m x^{m+n}$ is 0, forcing $a_n b_m = 0$. Then, consider the polynomial $a_n g(x) \in R[x]$: Since $a_n g(x) = a_n b_m x^m + \dots + a_n b_0$, with $a_n b_m = 0$, one either has $a_n g(x) = 0$, or $\deg(a_n g(x)) < m$. Yet, notice that $f(x) \cdot a_n g(x) = a_n (f(x)g(x)) = 0$, if $a_n g(x) \neq 0$, with its degree smaller than m (which is assumed to be minimal degree among all polynomials with $f(x)h(x) = 0$), this is a contradiction. So, one must have $a_n g(x) = 0$.

Now, inductively, given integer r with $0 \leq r \leq n$, suppose all index $0 \leq s < r$ satisfies the equation $a_{n-s} g(x) = 0$. Then, one has the following:

$$\begin{aligned} 0 = f(x)g(x) &= (a_n x^n + \dots + a_{n-(r-1)} x^{n-(r-1)})g(x) + (a_{n-r} x^{n-r} + \dots + a_0)g(x) \\ &= (a_{n-r} x^{n-r} + \dots + a_0)g(x) = (a_{n-r} x^{n-r} + \dots + a_0)(b_m x^m + \dots + b_0) \end{aligned} \quad (4.1)$$

This is due to the fact that $a_n g(x) = \dots = a_{n-(r-1)} g(x) = 0$ based on our induction hypothesis.

Then, the above equation provides that $a_{n-r} b_m = 0$. Hence, the term $a_{n-r} g(x) = a_{n-r} (b_m x^m + \dots + b_0) = a_{n-r} b_{m-1} x^{m-1} + \dots + a_{n-r} b_0$, satisfies either $a_{n-r} g(x) = 0$, or $\deg(a_{n-r} g(x)) < m$. However, if $a_{n-r} g(x) \neq 0$, then the term $f(x) \cdot a_{n-r} g(x) = a_{n-r} (f(x)g(x)) = 0$, while $\deg(a_{n-r} g(x)) < m$, this violates the minimality of m , which is a contradiction. Therefore, we once again has $a_{n-r} g(x) = 0$.

Finally, using induction we've proven that all integer $0 \leq r \leq n$ satisfies $a_{n-r} g(x) = 0$, then in particular all $a_{n-r} b_m = 0$ (the leading coefficient of $a_{n-r} g(x)$). Hence, $b_m f(x) = (a_n b_m) x^n + \dots + (a_0 b_m) = 0$, showing $b_m \neq 0$ in R is the desired element with $b_m f(x) = 0$.

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Problem 5

Prove that every PID is a UFD.

Solution: Let R be a PID. First, we'll show that all element must admit a factorization into products of irreducibles:

Suppose the contrary that R contains some nonzero element that doesn't admit a factorization, say a . Then, a must be reducible, there exists $a_0, a_1 \in R$, both nonzero non-units, such that $a = a_0 a_1$. Notice that one of the a_0, a_1 cannot admit a factorization, or else if both a_0, a_1 can be written

as finite product of irreducibles, so is $a = a_0 a_1$, which contradicts our assumption that a doesn't have a factorization into products of irreducibles. So, WLOG, say a_1 doesn't have a factorization.

Then, $a_1 = a_{10} a_{11}$ for some nonzero non-units, and the above claim also applies, one of a_{10}, a_{11} doesn't admit a factorization. WLOG, say a_{11} doesn't admit a factorization.

Inductively, one can keep factoring $a_{1\dots 1}$ (n ones) into $a_{1\dots 10} a_{1\dots 11}$, where both elements are nonzero non-units, and WLOG one can always say $a_{1\dots 11}$ doesn't admit a factorization.

Now, notice that this forms a chain of ideals:

$$(a) \subset (a_1) \subset (a_{11}) \subset \dots \subset (a_{1\dots 1}) \subset \dots \quad (5.1)$$

This is because each $a_{1\dots 1} = a_{1\dots 10} a_{1\dots 11}$.

Moreover, all the inclusions are proper: if there exists $(a_{1\dots 1}) = (a_{1\dots 11})$, this implies that $a_{1\dots 11} = b a_{1\dots 1}$ for some $b \in R$, which further implies $a_{1\dots 1} = a_{1\dots 10} a_{1\dots 11} = a_{1\dots 10} b a_{1\dots 1}$, or $a_{1\dots 1}(1 - a_{1\dots 10} b) = 0$. Yet, with the construction, $a_{1\dots 1} \neq 0$, with R being an integral domain, this enforces $1 - a_{1\dots 10} b = 0$, or $a_{1\dots 10} b = 1$. Yet, this is absurd, since $a_{1\dots 10}$ must be non-unit based on our construction. So, $(a_{1\dots 1}) \neq (a_{1\dots 11})$, the inclusion must be proper.

Then, here comes a contradiction:

Let $I_n := (a_{1\dots 1})$ (the element with n ones as index). Given the ascending chain of ideals $(a) \subset I_1 \subset I_2 \subset \dots \subset I_n \subset \dots$, define the set $J := \bigcup_{n \geq 1} I_n$, notice that J is an ideal:

- Given any $x, y \in J$, there exists $n, m \geq 1$, such that $x \in I_n$ and $y \in I_m$. WLOG, say $n \geq m$, then the ascending chain implies $I_m \subseteq I_n$, showing that $y \in I_n$ also. Therefore, the subtraction $x - y \in I_n \subseteq J$, showing J satisfies the subgroup test, which is a subgroup under addition.
- Furthermore, given any $r \in R$ and $x \in J$, there exists I_n such that $x \in I_n$, so $rx \in I_n \subseteq J$, showing J is an ideal.

Yet, by the PID property, there exists $d \in J$, such that $J = (d)$. So, there exists $k \geq 1$, such that $d \in I_k$, which implies $J \subseteq I_k \subset I_{k+1} \subset \dots \subset J$. Hence, all $n \geq k$ satisfies $I_n = J$, showing the chain is not strictly ascending, which contradicts the previous construction.

As a conclusion, all element must admit a factorization into products of irreducibles.

Now, we claim that every element has unique such factorization (up to associates): given any nonzero $a \in R$, let $a = p_1 \dots p_m = q_1 \dots q_n$ be two factorizations of a as products of irreducibles. Recall that in a PID, all irreducible elements are prime, hence with the equality showing $p_1 \mid q_1 \dots q_n$, the prime property of p_1 implies $p_1 \mid q_i$ for some index $1 \leq i \leq n$. Up to rearrangement, WLOG one can assume it's q_1 , hence $p_1 \mid q_1$; but, with q_1 being irreducible, if $q_1 = b p_1$ for some $b \in R$, b must be a unit (since p_1 is not a unit). So, $p_1 \sim q_1$; by multiplying suitable units to other irreducible factors, one can assume $p_1 = q_1$, so using the left cancellation property of integral domain, one gets $p_2 \dots p_m = q_2 \dots q_n$.

Finally, one can apply the above process to the new equality to derive $p_2 \sim q_2$ (by the prime property of p_2 and q_2), and inductively show that each $p_i \sim q_i$ for all index $1 \leq i \leq m$. This enforces $m = n$ (since one needs $m \leq n$ for all p_i to have an associate in the q_j 's; also, $m \geq n$, since after cancelling all p_i 's with their associate q_i 's, the left side of $p_1 \dots p_m = q_1 \dots q_n$ is left with unit, so the right side can't have any irreducibles left). Hence, one has $p_1 \dots p_m = q_1 \dots q_m$, with each $p_i \sim q_i$ up to rearrangement. This shows the factorization is unique up to associates.

Hence, PID has all nonzero elements admit a factorization, and is unique, showing a PID is a UFD.

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Problem 6

Let R be a UFD in which every nonzero prime ideal is maximal. Prove or disprove that R is a PID.

Solution: We'll prove that R is a PID, given all the conditions stated in the problem.

First, we'll prove that all prime ideals are principal ideals in R . Given $P \subseteq R$ a prime ideal, it's maximal by the assumption.

Then, notice that it must contain an irreducible element: Pick any nonzero $a \in P$, use the unique factorization property of R to write $a = p_1 \dots p_n$, products of irreducibles. Since $p_1 \dots p_n = a \in P$, by the prime property, either $p_1 \in P$ or $p_2 \dots p_n \in P$; if $p_1 \in P$ then we find an irreducible element in P , else if $p_2 \dots p_n \in P$, since its length as products of irreducibles decreases by 1, one can inductively deduce that at least one of the $p_i \in P$ (where $2 \leq i \leq n$).

WLOG, up to reordering, we can say $p_1 \in P$ using the above process. Then, the ideal $(p_1) \subseteq P$ is generated by irreducible element, which is a prime ideal in a UFD; furthermore, by our assumption all prime ideals are maximal, hence (p_1) is also a maximal ideal. With (p_1) , P both be maximal ideals, $(p_1) \subseteq P$ implies $(p_1) = P$, hence P is principal.

Now, we'll proceed using the following lemma:

Lemma

Given R an integral domain, then R is a PID iff all prime ideals are principal.

Proof:

$\Rightarrow$:

If R is a PID, then consequently all prime ideals must be principal.

$\Leftarrow$:

We'll approach the converse by contradiction. Suppose all prime ideals are principal, but R is not a PID. Let Σ denote the set collecting all the non-principal ideals in R , by the assumption $\Sigma \neq \emptyset$, and with inclusion as the partial order Σ forms a POset.

Now, let $C \subseteq \Sigma$ be a chain, we claim that C has an upper bound within Σ : Consider the ideal $I_C := \bigcup_{J \in C} J$, the union of all ideals in C . Let's check the ideal property of I_C as practice:

- Given any $x, y \in I_C$, there exists ideals $J_x, J_y \in C$, such that $x \in J_x$ and $y \in J_y$. Since C is a chain, either $J_x \subseteq J_y$ or $J_y \subseteq J_x$ (WLOG, say $J_x \subseteq J_y$), then one yields $x \in J_x \subseteq J_y$. By the fact that ideals are subgroup under addition, one has $x, y \in J_y$ implies $x - y \in J_y \subseteq I_C$, hence I_C satisfies the subgroup test (under addition), which is a subgroup under addition.
- Given any $x \in I_C$ and $r \in R$, there exists ideal $J \in C$, such that $x \in J$. So, $rx \in J \subseteq I_C$, showing I_C is an ideal.

Notice that I_C cannot be a principal ideal, since if it's principal, say $I_C = (d)$ for some $d \in I_C$, then there exists an ideal $J \in C$, such that $d \in J$, showing $(d) \subseteq J \subseteq I_C = (d)$, or $J = (d)$. Yet, this contradicts the fact that $J \in \Sigma$ (or J is non-principal), so I_C can't be a principal ideal. Hence, $I_C \in \Sigma$, which also serves as an upper bound to the chain C .

Since all chain in Σ as a POset has an upper bound, apply Zorn's Lemma, Σ must contain some maximal element, say $I \in \Sigma$. Because I is non-principal, it's also not prime (by assumption, all prime ideals are principal). Thus, there exists $a, b \in R \setminus I$, such that $ab \in I$.

Now, let $I_a := I + (a)$, and $I_b := I + (b)$. Since $a, b \notin I$ by our choice, I_a, I_b are ideals strictly containing I . Which, by the maximality of $I \in \Sigma$, I_a, I_b are not in Σ , hence they're principal. In particular, $I_a = (\alpha)$ for some $\alpha \in I_a$.

Then, let's take the set $J := \{x \in R \mid xI_a \subseteq I\}$ (I believe in Atiyah Macdonald, this is called *Ideal Quotient*), which J is also an ideal:

- Given $x, y \in J$, since $xI_a, yI_a \subseteq I$, then $(x - y)I_a \subseteq I$, showing $x - y \in J$. Thus, J is a subgroup under addition (as it passes the subgroup test).
- Given $r \in R$ and $x \in J$, since $xI_a \subseteq I$, one has $(rx)I_a = r(xI_a) \subseteq I$, so $rx \in J$, showing J is an ideal.

Notice that by definition of an ideal, $I \subseteq J$ (since all $x \in I$ satisfies $xI_a \subseteq I$); on the other hand, since $b \notin I$ satisfies $bI_a = b(I + (a)) = bI + b(a) = bI + (ab) \subseteq I$ (since $ab \in I$), one has $b \in J$, so J strictly contains I . By similar reasoning, $J = (\gamma)$ for some $\gamma \in J$.

Finally, given any $x \in I$, since it's strictly contained in $I_a = (\alpha)$, one has $x = s_\alpha \alpha$ for some $s_\alpha \in R$; then, all $r \in R$ has satisfied $s_\alpha \cdot r\alpha = r \cdot s_\alpha \alpha = rx \in I$, showing that $s_\alpha(\alpha) = s_\alpha I_a \subseteq I$, so $s_\alpha \in J = (\gamma)$. Hence, there exists $s \in R$, such that $s_\alpha = s\gamma$, showing that $x = s_\alpha \alpha = s\gamma\alpha$, or $x \in (\alpha\gamma)$. Thus, $I \subseteq (\alpha\gamma)$.

On the other hand, let's not forget that $\gamma \in J$, so with $\alpha \in I_a$, one has $\alpha\gamma \in I$, hence $(\alpha\gamma) \subseteq I$. This concludes that $I = (\alpha\gamma)$, or I is a principal ideal. Yet, this contradicts the fact that $I \in \Sigma$ (showing I is non-principal).

Therefore, the assumption is wrong, R must be a PID. $\square$

Finally, using the above lemma, because all prime ideals of R is proven to be principal, R must be a PID.

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Problem 7

Prove or disprove that $\mathbb{Z}[x]/(x^2 + 5)$ is a PID.

Solution: We'll prove that the given ring is not a PID, by showing that it's not a UFD (since all PID's are UFD's, which is proven in **Problem 5**).

Here we'll implicitly use the observation that $\mathbb{Z}[x]/(x^2 + 5) \cong \mathbb{Z}[\sqrt{-5}]$.

Notice first that $\overline{x^2} = -5$, and $\overline{x^3} = \overline{x^2} \cdot \overline{x} = -5\overline{x}$ in this ring, which are integer multiples of 1 or $\overline{x}$; also, notice that for any monomials $\overline{x^k}$ with $k \geq 3$, it can be written as $\overline{x^2} \cdot \overline{x^{k-2}} = -5\overline{x^{k-2}}$ (which is the quotient of a smaller degree). Hence, inductively all monomials with degree $k \geq 2$ has a quotient that can be an integer multiples of 1 or $\overline{x}$, showing that all polynomials $f(x) \in \mathbb{Z}[x]$ has its quotient $\overline{f(x)} \in \mathbb{Z}[x]/(x^2 + 5)$ being an integer combination of 1 and $\overline{x}$.

Moreover, such integer combination of 1 and $\overline{x}$ must be unique: Suppose $\overline{f(x)} = a + b\overline{x} = a' + b'\overline{x}$, then one has $(a - a') + (b - b')\overline{x} = 0$, showing that $(a - a') + (b - b')x \in (x^2 + 5)$. So, either $(a - a') + (b - b')x = 0$, or there exists nonzero $g(x) \in \mathbb{Z}[x]$, satisfying $g(x)(x^2 + 5) = (a - a') + (b - b')x$. Yet, the second possibility leads to a contradiction, due to the fact that $\mathbb{Z}[x]$ is an integral domain, so $\deg(g(x)(x^2 + 5)) \geq \deg(x^2 + 5) = 2 > 1 = \deg((a - a') + (b - b')x)$, showing the degrees don't match. Hence, one must have $(a - a') + (b - b')x = 0$, or $a = a'$ and $b = b'$. This shows the unique integer combination part.

Now, define a function $N : \mathbb{Z}[x]/(x^2 + 5) \rightarrow \mathbb{Z}_{\geq 0}$ by $N(a + b\overline{x}) = a^2 + 5b^2$. This function is well-defined, since all $f(x) \in \mathbb{Z}[x]$ has unique $a, b \in \mathbb{Z}$, such that $\overline{f(x)} = a + b\overline{x}$ in $\mathbb{Z}[x]/(x^2 + 5)$.

Then, notice that this function N has the following properties:

1. Given any $a, b, c, d \in \mathbb{Z}$, the multiplication $(a + b\overline{x})(c + d\overline{x}) = ac + (bc + ad)\overline{x} + bd\overline{x^2} = (ac - 5bd) + (bc + ad)\overline{x}$. Hence, the function N satisfies the following equality:

$$\begin{aligned}
 N((a + b\overline{x})(c + d\overline{x})) &= (ac - 5bd)^2 + 5(bc + ad)^2 \\
 &= a^2c^2 - 10abcd + 5b^2d^2 + 5b^2c^2 + 10abcd + 5a^2d^2 \\
 &= a^2c^2 + 5b^2d^2 + 5b^2c^2 + 5a^2d^2 \\
 &= a^2(c^2 + 5d^2) + 5b^2(c^2 + 5d^2) \\
 &= (a^2 + 5b^2)(c^2 + 5d^2) \\
 &= N(a + b\overline{x}) \cdot N(c + d\overline{x})
 \end{aligned} \tag{7.1}$$

This shows the multiplicative property of N .

2. Based on 1 above, the only units are ± 1 : It's clear why these two elements are units, to argue they're the only units, suppose $a + b\overline{x} \in \mathbb{Z}[x]/(x^2 + 5)$ is also a unit, there exists $c + d\overline{x} \in \mathbb{Z}[x]/(x^2 + 5)$, such that $(a + b\overline{x})(c + d\overline{x}) = 1$. Plug in to the function N , one yields:

$$1 = 1^0 + 5 \cdot 0^2 = N(1 + 0\overline{x}) = N(1) = N((a + b\overline{x})(c + d\overline{x})) = N(a + b\overline{x})N(c + d\overline{x}) \tag{7.2}$$

This shows that both $N(a + b\overline{x}) = a^2 + 5b^2$, and $N(c + d\overline{x}) = c^2 + 5d^2$ must divide 1, which the values can only be ± 1 ; furthermore, these two values are nonnegative, since arbitrary integer x, y has $x^2, y^2 \geq 0$, so $x^2 + 5y^2 \geq 0$ also, forcing both values to be 1. Finally, this implies that $b = 0$, since if $b \neq 0$, $b^2 \geq 1$, showing that $a^2 + 5b^2 \geq 5b^2 \geq 5 > 1$, which the value $N(a + b\overline{x}) \neq 1$, showing a contradiction.

As a result, $a + b\overline{x} = a$, which it satisfies $N(a + b\overline{x}) = a^2 + 5b^2 = a^2 = 1$, so $a = \pm 1$, showing ± 1 are the only units in this ring.

Also, this shows that $N(a + b\bar{x}) = a^2 + 5b^2 = 1$ implies $a + b\bar{x} = \pm 1$, since if $b \neq 0$ the above equality shows $a^2 + 5b^2 > 1$, so $b = 0$; as a result, one has $a^2 = 1$, or $a = \pm 1$. Hence, $a + b\bar{x} = \pm 1$. So, one also concludes that $N(a + b\bar{x}) = 1$ iff $a + b\bar{x} = \pm 1$ iff $a + b\bar{x}$ is a unit.

3. Also based on 1 above, $\mathbb{Z}[x]/(x^2 + 5)$ is an integral domain: Since $N(0) = 0^2 + 5 \cdot 0^2 = 0$, suppose $a + b\bar{x}, c + d\bar{x} \in \mathbb{Z}[x]/(x^2 + 5)$ satisfies $(a + b\bar{x})(c + d\bar{x}) = 0$, then one has:

$$0 = N(0) = N((a + b\bar{x})(c + d\bar{x})) = N(a + b\bar{x}) \cdot N(c + d\bar{x}) = (a^2 + 5b^2)(c^2 + 5d^2) \quad (7.3)$$

Because $\mathbb{Z}$ is an integral domain, the above equality forces $a^2 + 5b^2 = 0$ or $c^2 + 5d^2 = 0$. However, given any $x, y \in \mathbb{Z}$, if $x^2 + 5y^2 = 0$, it enforces both $x, y = 0$ (Suppose $y \neq 0$, then $y^2 \geq 1$, so $x^2 + 5y^2 \geq 5y^2 \geq 5 > 0$; likewise, if $x \neq 0$, then $x^2 + 5y^2 \geq x^2 > 0$). So, if $a^2 + 5b^2 = 0$, then $a, b = 0$, forcing $a + b\bar{x} = 0$; similarly, if $c^2 + 5d^2 = 0$, one has $c + d\bar{x} = 0$ instead.

So, $\mathbb{Z}[x]/(x^2 + 5)$ is an integral domain.

Then, using these facts we claim that $2, 3, 1 + \bar{x}, 1 - \bar{x} \in \mathbb{Z}[x]/(x^2 + 5)$ are irreducible, and non-associates to each other:

4. Given that $N(2) = 4$, $N(3) = 9$, and $N(1 + \bar{x}) = N(1 - \bar{x}) = 1 + 5 = 6$. Notice that if $a + b\bar{x}, c + d\bar{x} \in \mathbb{Z}[x]/(x^2 + 5)$ are associates, there exists unit (namely 1 or -1) such that $c + d\bar{x} = \pm(a + b\bar{x})$ (the sign dependent on the situation), which implies that $N(c + d\bar{x}) = N((\pm 1) \cdot (a + b\bar{x})) = N(\pm 1) \cdot N(a + b\bar{x}) = N(a + b\bar{x})$.

So, associates must have the same values under N , showing 2, and 3, and $(1 + \bar{x}), (1 - \bar{x})$ must all belong to distinct associates class (because $N(2) = 4$, $N(3) = 9$, $N(1 + \bar{x}) = N(1 - \bar{x}) = 6$ are all distinct); furthermore, $1 + \bar{x}, 1 - \bar{x}$ are not associates, simply because multiplying by 1 or -1 on one of them doesn't recover the other element.

5. To prove they're irreducibles, we'll show there doesn't exist element $a + b\bar{x}$, such that $N(a + b\bar{x}) = 2$ or 3 (the only nonnegative integers possibly dividing $N(2) = 4$, $N(3) = 9$, or $N(1 + \bar{x}) = N(1 - \bar{x}) = 6$).

Suppose there exists $a + b\bar{x}$ satisfying $N(a + b\bar{x}) = a^2 + 5b^2 = 2$ or 3. First notice that $b = 0$ (since if $b \neq 0$, $b^2 \geq 1$, one has $a^2 + 5b^2 \geq 5b^2 \geq 5 > 2, 3$, which forms a contradiction). So, it results in $a^2 = 2$ or 3. Yet, both of these are prime numbers, which cannot be square of some integers, which we reach a contradiction. Therefore, there doesn't exist element with $N(a + b\bar{x}) = 2$ or 3.

Now, notice that this verifies $2, 3, 1 + \bar{x}, 1 - \bar{x}$ are all irreducibles:

Suppose the contrary that (for instance) 2 is reducible, then $2 = (a + b\bar{x})(c + d\bar{x})$ for $a + b\bar{x}, c + d\bar{x}$ being non-units. Which, the condition proven before shows that $N(a + b\bar{x}), N(c + d\bar{x}) \neq 1$. With $4 = N(2) = N(a + b\bar{x})N(c + d\bar{x})$, these two values are nonnegative, non-1 divisors of 4, which must be 2. Yet, this contradicts the result proven above, where there's no element with $N(a + b\bar{x}) = 2$. So, 2 must be irreducible.

Likewise, the same proof applies for 3, since $N(3) = 9$, and there doesn't exist element with $N(a + b\bar{x}) = 3$.

Finally, for $1 + \bar{x}, 1 - \bar{x}$, if one of them are reducible, say $1 + \bar{x} = (a + b\bar{x})(c + d\bar{x})$ for two non-units. The above equality states that $6 = N(1 + \bar{x}) = N(a + b\bar{x})N(c + d\bar{x})$, so $N(a + b\bar{x}), N(c + d\bar{x})$ are non-1 divisors of 6, showing one must be 2 and the other one must be 3. But this is again a contradiction based on the same reasoning above.

Thus, we can conclude that $2, 3, 1 - \bar{x}, 1 + \bar{x}$ are irreducibles, and are non-associates of each other.

Finally, this shows that $\mathbb{Z}[x]/(x^2 + 5)$ is not a PID, because it's not a UFD: Consider $6 = 2 \cdot 3$, and $6 = 1 + 5 = 1 - \bar{x}^2 = (1 - \bar{x})(1 + \bar{x})$. With $2, 3, 1 + \bar{x}, 1 - \bar{x}$ be non-associate irreducibles, 6 has two distinct factorizations, showing the given ring is not a UFD.

8 D

Problem 8

Let R be a commutative ring and $a, b \in R$ be such that $a \in \text{Nil}(R)$ and $b \in U(R)$. Prove or disprove that $a + b \in U(R)$.

Solution: We'll prove that $b + a \in U(R)$.

Given b is a unit and a is nilpotent, there exists $n > 0$, such that $a^n = 0$. Also, one can rewrite $b + a = b(1 + b^{-1}a)$, which $b^{-1}a$ is also nilpotent (since $(b^{-1}a)^n = b^{-n}a^n = 0$ due to commutativity of R).

Now, we claim that $1 + b^{-1}a$ is a unit in R : Consider the term $\sum_{i=0}^{n-1} (-b^{-1}a)^i$ (where $(-b^{-1}a)^0 = 1$ by convention), one has the following:

$$\begin{aligned}
 (1 + b^{-1}a) \left(\sum_{i=0}^{n-1} (-b^{-1}a)^i \right) &= \sum_{i=0}^{n-1} (-b^{-1}a)^i - (-b^{-1}a) \sum_{i=0}^{n-1} (-b^{-1}a)^i \\
 &= \sum_{i=0}^{n-1} (-b^{-1}a)^i - \sum_{i=1}^n (-b^{-1}a)^i \\
 &= (-b^{-1}a)^0 - (-b^{-1}a)^n \\
 &= 1 - 0 = 1
 \end{aligned} \tag{8.1}$$

So, one has the following:

$$(b + a) \left(b^{-1} \sum_{i=0}^{n-1} (-b^{-1}a)^i \right) = b(1 + b^{-1}a)b^{-1} \left(\sum_{i=0}^{n-1} (-b^{-1}a)^i \right) = 1 \tag{8.2}$$

This shows that $(a + b) \in U(R)$.

9 D

Problem 9

Let R be a commutative ring and $f(x) \in R[x]$. Prove or disprove that $f(x) \in \text{Nil}(R[x])$ iff all coefficients of $f(x)$ lie in $\text{Nil}(R)$.

Solution: We'll prove the equivalence between the two statements.

$\Rightarrow$:

First, suppose $f(x) = a_0 + a_1x + \dots + a_nx^n \in \text{Nil}(R[x])$. We'll prove by induction on n , that all of its coefficient is in $\text{Nil}(R)$.

For the case $n = 0$, it's clear that $f(x) = a_0$ is nilpotent implies there exists some $k > 0$, such that $f(x)^k = a_0^k = 0$, showing $a_0 \in \text{Nil}(R)$.

Now, suppose given n , one has $f(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} \in \text{Nil}(R[x])$ implies each coefficient $a_i \in \text{Nil}(R)$. Given now that $f(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n \in \text{Nil}(R[x])$, there exists $k > 0$, such that $f(x)^k = 0$. Notice that the highest degree term in $f(x)^k$ is given by $(a_nx^n)^k = a_n^kx^{nk}$, which $f(x)^k = 0$ implies that $a_n^k = 0$, showing $a_n \in \text{Nil}(R)$.

However, this also implies that $(a_nx^n)^k = a_n^kx^{nk} = 0$, showing $a_nx^n \in \text{Nil}(R[x])$, so the term $f(x) - a_nx^n = a_0 + a_1x + \dots + a_{n-1}x^{n-1} \in \text{Nil}(R[x])$ (since the nilradical is an ideal). By our induction hypothesis, one has $a_0, a_1, \dots, a_{n-1} \in \text{Nil}(R)$, showing all coefficients of $f(x)$ is in $\text{Nil}(R)$.

Hence, by induction one concludes that $f(x) \in \text{Nil}(R[x])$ implies all its coefficients are in $\text{Nil}(R)$.

$\Leftarrow$:

If $f(x) = a_0 + a_1x + \dots + a_nx^n$ has all coefficients lie in $\text{Nil}(R)$, for each index $k \in \{0, 1, \dots, n\}$, there exists $l_k > 0$, such that $a_k^{l_k} = 0$. As a result, a_kx^k satisfies $(a_kx^k)^{l_k} = a_k^{l_k}x^{kl_k} = 0$, showing that $a_kx^k \in \text{Nil}(R[x])$.

Then, since $f(x)$ is a finite sum of nilpotent elements (since each monomial is verified to be nilpotent), $f(x) \in \text{Nil}(R[x])$ (because nilradical is an ideal).

The above two proofs verifies the equivalence of the two statements.

10 D

Problem 10

Let R be a commutative ring and $f(x) \in R[[x]]$ be of the form $f(x) = \sum_{i \geq 0} a_i x^i$. Prove or disprove that $f(x) \in U(R[[x]]) \iff a_0 \in U(R)$.

Solution: We'll prove the equivalence $f(x) \in U(R[[x]]) \iff a_0 \in U(R)$.

$\implies$:

First, suppose $f(x) = \sum_{i \geq 0} a_i x^i \in U(R[[x]])$, then there exists $g(x) = \sum_{j \geq 0} b_j x^j \in R[[x]]$, such that $f(x)g(x) = 1$. Expand the product, one has the following:

$$1 = f(x)g(x) = \left(\sum_{i \geq 0} a_i x^i \right) \left(\sum_{j \geq 0} b_j x^j \right) = \sum_{k \geq 0} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k \quad (10.1)$$

So, consider the constant term, the left hand side provides 1, while the right hand side provides $\sum_{i=0}^0 a_i b_{0-i} = a_0 b_0$. Hence, one concludes that $a_0 b_0 = 1$, or $a_0 \in U(R)$.

$\impliedby$:

Now, suppose that $a_0 \in U(R)$, consider the following formula:

Define $g(x) = \sum_{j \geq 0} b_j x^j$ by $b_0 := a_0^{-1}$, and each $k > 0$ has $b_k := -\sum_{i=1}^k a_0^{-1} a_i b_{k-i}$. Then, we have the following:

$$f(x)g(x) = \left(\sum_{i \geq 0} a_i x^i \right) \left(\sum_{j \geq 0} b_j x^j \right) = \sum_{k \geq 0} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k \quad (10.2)$$

Which, for $k = 0$, the coefficient is $a_0 b_0 = a_0 a_0^{-1} = 1$; for $k = 1$, one has coefficient $a_1 b_0 + a_0 b_1 = a_1 b_0 + a_0 (-a_0^{-1} a_1 b_0) = 0$. And for all $k \geq 1$, one has the following by the definition of b_k :

$$\begin{aligned} \sum_{i=0}^k a_i b_{k-i} &= a_0 b_k + \sum_{i=1}^k a_i b_{k-i} = a_0 \left(-\sum_{i=1}^k a_0^{-1} a_i b_{k-i} \right) + \sum_{i=1}^k a_i b_{k-i} \\ &= -\sum_{i=1}^k a_i b_{k-i} + \sum_{i=1}^k a_i b_{k-i} = 0 \end{aligned} \quad (10.3)$$

So, all $k \geq 1$ has the coefficient of $f(x)g(x)$ at degree k being $\sum_{i=0}^k a_i b_{k-i} = 0$, showing $f(x)g(x)$ has constant coefficient 1, and all non-constant coefficient 0, showing $f(x)$ has an inverse, hence a unit in $R[[x]]$.

11 D

Problem 11

Let R be a commutative ring and $f(x) \in R[[x]]$. Suppose $f(x) \in \text{Nil}(R[[x]])$. Prove or disprove that all coefficients of $f(x)$ are nilpotent in R .

Solution: We'll prove that all coefficients of $f(x)$ are all nilpotent in R via induction.

Let $f(x) = \sum_{n \geq 0} a_n x^n$. Since $f(x)$ is nilpotent, there exists $k > 0$, such that $f(x)^k = 0$.

So, for $n = 0$, since $f(x)^k$ has constant term a_0^k , this enforces $a_0^k = 0$, hence a_0 is nilpotent in R .

Now, suppose for given n , one has $a_0, \dots, a_{n-1} \in R$ all being nilpotent. Then, as a result each index $0 \leq i < n$ has $a_i x^i \in R[[x]]$ being nilpotent also (since the only potential nonzero coefficient is nilpotent). Thus, one has $\sum_{m \geq n} a_m x^m = \sum_{m \geq 0} a_m x^m - (a_0 + a_1 x + \dots + a_{n-1} x^{n-1}) = f(x) - (a_0 + a_1 x + \dots + a_{n-1} x^{n-1}) \in \text{Nil}(R[[x]])$ (since nilradical is an ideal).

So, there exists $l_n > 0$, such that $\left(\sum_{m \geq n} a_m x^m\right)^{l_n} = 0$. Thus, one derives the following:

$$0 = \left(\sum_{m \geq n} a_m x^m\right)^{l_n} = \left(x^n \sum_{m \geq n} a_m x^{m-n}\right)^{l_n} = \left(x^n \sum_{m \geq 0} a_{n+m} x^m\right)^{l_n} = x^{n l_n} (a_n^{l_n} + \dots) \quad (11.1)$$

Which, the smallest degree term $x^{n l_n}$ has coefficient $a_n^{l_n}$, showing that $a_n^{l_n} = 0$, or $a_n \in R$ is nilpotent.

Hence, by induction, one can conclude that all coefficients of $f(x)$ are nilpotent in R .

12 D

Problem 12

Let R be a commutative ring and $f(x) = \sum_{i \geq 0} a_i x^i \in R[[x]]$. Prove or disprove that $f(x) \in J(R[[x]]) \iff a_0 \in J(R)$.

Solution: We'll prove that $f(x) \in J(R[[x]]) \iff a_0 \in J(R)$.

First, recall the characterization of Jacobson radical: Given a commutative ring S , one has $a \in J(S)$ iff all $t \in S$ satisfies $1 - at$ being a unit.

Second, we now know that $f(x) \in R[[x]]$ is a unit, iff its constant term $a_0 \in R$ is a unit (done in **Problem 10**).

$\implies$:

Suppose $f(x) \in J(R[[x]])$, then given $f(x) = \sum_{i \geq 0} a_i x^i$, since all $g(x) \in R[[x]]$ satisfies $1 - g(x)f(x) \in U(R[[x]])$, in particular, given any $b \in R \hookrightarrow R[[x]]$, one has $1 - bf(x)$ being a unit. However, it has the following form:

$$1 - bf(x) = 1 - \sum_{i \geq 0} ba_i x^i = (1 - ba_0) + \sum_{i \geq 1} ba_i x^i \quad (12.1)$$

Which, its constant term is $1 - ba_0$. Based on the second statement listed beforehand, $1 - bf(x)$ is a unit iff its constant term $1 - ba_0$ is a unit. So, all $b \in R$ has $1 - ba_0$ being a unit, showing $a_0 \in J(R)$.

$\impliedby$:

Suppose $a_0 \in J(R)$, then all $b \in R$ satisfies $1 - ba_0 \in U(R)$. Now, given any $g(x) = \sum_{j \geq 0} b_j x^j \in R[[x]]$, one has $1 - g(x)f(x)$ being the following:

$$\begin{aligned} 1 - g(x)f(x) &= 1 - \left(\sum_{j \geq 0} b_j x^j \right) \left(\sum_{i \geq 0} a_i x^i \right) \\ &= 1 - \sum_{k \geq 0} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k \\ &= (1 - a_0 b_0) + \sum_{k \geq 1} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k \end{aligned} \quad (12.2)$$

So, $1 - g(x)f(x)$ has its constant term $1 - a_0 b_0$ being a unit, showing $1 - g(x)f(x)$ is a unit. Since such relation is true for all $g(x) \in R[[x]]$, $f(x) \in J(R[[x]])$.

13 D

Problem 13

Let R be a commutative ring in which every element x satisfies $x^n = x$ (n depends on x). Prove or disprove that every prime ideal of R is maximal.

Solution: Here we'll prove by contradiction that every prime ideal of R is maximal.

Suppose there exists prime ideal $P \subsetneq R$ that is not maximal, then the quotient ring R/P is only an integral domain, but not a field. Thus, there exists an element $x \in R$, such that its quotient $\bar{x} \in R/P$ is nonzero, but not a unit.

Now, since there exists $n \in \mathbb{N}$ such that $x^n = x$ (in particular, one can choose $n > 1$), it's the same for the quotient (where $\bar{x}^n = \overline{x^n} = \bar{x}$). So, since $\bar{x}$ is nonzero and R/P is an integral domain, $\bar{x}^{n-1} \neq 0$; on the other hand, since $\bar{x}$ is not a unit either, $\bar{x}^{n-1} \neq 1$.

However, consider the following equality:

$$1 \cdot \bar{x} = \bar{x} = \bar{x}^n = \bar{x}^{n-1} \cdot \bar{x} \tag{13.1}$$

Hence, one gets that $(1 - \bar{x}^{n-1}) \cdot \bar{x} = 0$. But, recall that $\bar{x}^{n-1} \neq 1$ (so $1 - \bar{x}^{n-1} \neq 0$), while $\bar{x} \neq 0$ also, showing that $\bar{x}$ is in fact a nonzero zero-divisor. Yet, this contradicts the assumption that R/P is an integral domain.

So, our assumption is false, every prime ideal in R must be maximal.

14 D

Problem 14

Let R be a commutative ring such that $R[x]$ is a PID. Prove or disprove that R is a field.

Solution: We'll prove that R is a field.

Given any nonzero element $a \in R$, consider the following ideal $(a, x) \subseteq R[x]$: Since $R[x]$ is a PID, one has $(a, x) = (f(x))$ for some $f(x) \in R[x]$. Then, since $a = g(x)f(x)$ for some $g(x) \in R[x]$, using the degree formula of integral domain polynomial ring, we get:

$$0 = \deg(a) = \deg(g(x)f(x)) = \deg(g(x)) + \deg(f(x)) \quad (14.1)$$

So, with the degrees being nonnegative, one has $\deg(f(x)) = \deg(g(x)) = 0$, showing there exists nonzero $b, c \in R$, such that $f(x) = b$, and $g(x) = c$.

On the other hand, since $b = f(x) = k(x)a + l(x)x$ for some $k(x), l(x) \in R[x]$ (by the ideal relation), take the evaluation map $\varphi : R[x] \rightarrow R$ by $\varphi(x) = 0$, one has the following:

$$b = \varphi(b) = \varphi(k(x)a + l(x)x) = k(0)a + l(0)0 = k(0)a \quad (14.2)$$

Combining with the previous relation $a = f(x)g(x) = bc$, one has $a = c \cdot k(0)a$, so $a(1 - c \cdot k(0)) = 0$; with the assumption $a \neq 0$, one must have $1 = c \cdot k(0)$, or c is a unit.

Hence, $b = c^{-1}a$, showing that a, b are associates, implying $(a) = (b)$, or $(a, x) = (b, x) = (f(x)) = (b) = (a)$.

Finally, since $x \in (a)$, there exists $r(x) \in R[x]$, such that $r(x)a = x$. Since $\deg(x) = 1$, this enforces $\deg(r(x)a) = \deg(r(x)) + \deg(a) = 1$; with $\deg(a) = 0$, then $\deg(r(x)) = 1$, so there exists $p, q \in R$, such that $r(x) = px + q$.

However, with $x = r(x)a = (ap)x + (aq)$, this implies $ap = 1$, showing a is a unit.

Hence, all nonzero element $a \in R$ is a unit, showing R is a field.

15 D

Problem 15

Let R be a commutative ring. Prove or disprove that $\text{Nil}(R[x]) = J(R[x])$.

Solution: We'll prove that $\text{Nil}(R[x]) = J(R[x])$.

First, it's clear that $\text{Nil}(R[x]) \subseteq J(R[x])$. Now, for the other inclusion, we require the following statement:

Lemma

$f(x) = a_0 + a_1x + \dots + a_nx^n \in R[x]$ is a unit $\iff a_0$ is a unit, and $a_1, \dots, a_n$ are nilpotent.

Proof:

$\Leftarrow$:

First, if a_0 is a unit and $a_1, \dots, a_n$ are nilpotent in R , each index $1 \leq i \leq n$ has $a_i x^i$ be nilpotent also. Hence, $f(x) = a_0 + (a_1 x + \dots + a_n x^n)$, where $a_1 x + \dots + a_n x^n \in \text{Nil}(R[x])$, and $a_0 \in U(R[x])$. Using **Problem 8**, $f(x)$ is a unit plus a nilpotent element, hence a unit in $R[x]$.

$\Rightarrow$:

Suppose $f(x) = a_0 + a_1 x + \dots + a_n x^n$ is a unit. Then, there exists $g(x) = b_0 + b_1 x + \dots + b_m x^m$, such that $f(x)g(x) = 1$. Notice that the left hand side has constant term being $a_0 b_0$, so $a_0 b_0 = 1$, showing that a_0 is a unit.

Now, to show each $a_1, \dots, a_n$ are nilpotent, we'll prove that they're contained in all prime ideals of R . Given any prime ideal $P \subseteq R$, recall that the extension of ideal $P^e \subseteq R[x]$ satisfies $R[x]/P^e \cong (R/P)[x]$. Hence, take the projection $\varphi : R[x] \twoheadrightarrow \frac{R[x]}{P^e} \cong (R/P)[x]$, one can interpret it as follow:

$$\varphi(r_0 + r_1 x + \dots + r_k x^k) = \bar{r}_0 + \bar{r}_1 x + \dots + \bar{r}_k x^k \quad (15.1)$$

where $\bar{r}$ represents the quotient of r in R/P .

Notice that R/P is an integral domain, so is $(R/P)[x]$, which implies that $U(R/P) = U((R/P)[x])$ (here's the proof): Given the inclusion $R/P \hookrightarrow (R/P)[x]$, it is clear that $U(R/P) \subseteq U((R/P)[x])$. Now, suppose $f(x) \in (R/P)[x]$ is a unit, then there exists $g(x) \in (R/P)[x]$, such that $f(x)g(x) = 1$. Recall that for integral domain polynomial ring, one has the following:

$$0 = \deg(1) = \deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x)) \quad (15.2)$$

Since degree is nonnegative, this enforces $\deg(f(x)) = \deg(g(x)) = 0$, showing both $f(x), g(x)$ is in fact a constant, hence in R/P . As a result, $f(x), g(x) \in U(R/P)$, showing $U((R/P)[x]) \subseteq U(R/P)$.

Finally, if consider $\varphi(f(x)) = \bar{a}_0 + \bar{a}_1 x + \dots + \bar{a}_n x^n$ in $(R/P)[x]$, it remains as a unit. Hence by the above statement, it enforces $\varphi(f(x))$ to be constant, showing that $\bar{a}_1, \dots, \bar{a}_n = 0$ in R/P , or $a_1, \dots, a_n \in P$. Then, since P is arbitrary prime ideal of R , $a_1, \dots, a_n$ lies in the intersection of all prime ideal, which is precisely $\text{Nil}(R)$. Thus, $a_1, \dots, a_n$ are nilpotent, and it finishes the lemma. $\square$

Back to the original problem, for any $f(x) = a_0 + a_1 x + \dots + a_n x^n \in J(R[x])$, every $g(x) \in R[x]$ satisfies $1 - g(x)f(x)$ is a unit. In particular, take $g(x) = x$, one has the following being a unit:

$$1 - xf(x) = 1 - x(a_0 + a_1 x + \dots + a_n x^n) = 1 - a_0 x - a_1 x^2 - \dots - a_n x^{n+1} \quad (15.3)$$

Hence, by the lemma, all the non-constant term coefficients are nilpotent, showing $-a_0, -a_1, \dots, -a_n$ are nilpotent, hence $a_0, a_1, \dots, a_n$ are all nilpotent.

Finally, using the statement proven in **Problem 11**, $f(x) \in \text{Nil}(R[x])$ iff all its coefficients are nilpotent in R . So, with all coefficients of $f(x)$ being nilpotent, $f(x) \in \text{Nil}(R[x])$. This shows the inclusion $J(R[x]) \subseteq \text{Nil}(R[x])$.

Hence, $\text{Nil}(R[x]) = J(R[x])$.